THE CATHOLIC DIOCESE OF TANGA

TANGA CATHOLIC SECONDARY SCHOOLS UNION (TACASSU)

FORM FOUR JOINT PRE-NATIONAL EXAMINATION 1

036 INFORMATION AND COMPUTER STUDIES

Time: 3 Hours **2026**

Instruction

1. This paper consists of sections **A, B** and **C** with total of **eleven** questions.
2. Answer **all** questions in section **A** and **B**, and **two** questions from section **C**.
3. Section **A** carries **16** marks; section **B 54** marks and section **C** carries **30** marks.
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room.

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SECTION A (16 MARKS)

Answer **all** questions in this section.

1. For each of the following items (i) to (x) choose the correct answer and write its letter in the box provided

(i) A school computer lab has several computers that need to share a printer and exchange files without using the Internet. Which arrangement would best meet this need?

A. Stand-alone computers
B. Computer network
C. Word-processing system
D. Database system
E. Multimedia system

(ii) A student accidentally deletes an important folder containing examination records. Which practice would have provided the best protection against permanent loss of the records?

A. Increasing screen brightness
B. Changing the desktop background
C. Creating regular backups
D. Installing a larger monitor
E. Formatting the keyboard

(iii) A school secretary wants to prepare a letter, apply different fonts, insert a school logo and print several copies. Which type of application software is most appropriate?

A. Spreadsheet software
B. Database software
C. Word-processing software
D. Presentation software
E. Antivirus software

(iv) The following are examples of structured tags in html, except;

A. <p>
B. <a>
C. <img>
D.
E. <form>

(v) Information is best defined as:

A. Raw facts collected from different sources
B. Processed data that is meaningful to users
C. Unorganized figures and symbols
D. Files stored in a computer
E. Computer instructions

(vi) A student prepared the following spreadsheet:

A	**B**
Marks	Grade
75	?

Which formula will correctly display **PASS** when the mark is 50 or above and **FAIL** otherwise?

A. =IF(A2>=50,"PASS","FAIL")
B. =IF(A2>50,"FAIL","PASS")
C. =IF(A2<=50,"PASS","FAIL")
D. =IF(A2=50,"FAIL","PASS")
E. =IF(A2<>50,"PASS","FAIL")

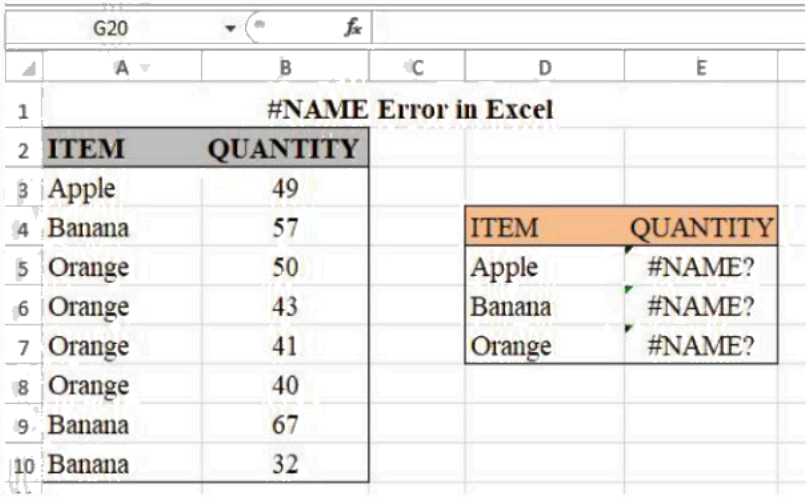
(vii) The process of rearranging files and data in the storage medium such as Solid-State Disk (SSD) is called?

A. Partitioning
B. Compression
C. Defragmentation
D. Back – up
E. Conversion

(viii) Grouping two or more adjacent cells into one large cell is referred to as

A. Grouping cells
B. Joining cells
C. Linking cells
D. Merging cells
E. Combining cells

(ix) The following spreadsheet was prepared to show the results of the students who did one different task. In cells E5 an error occurred. What might be the reason?



G20

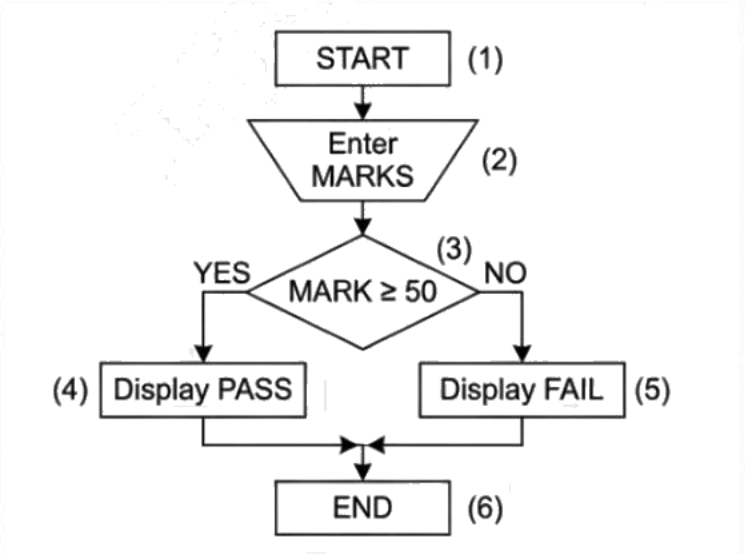
	A	B	C	D	E
1	#NAME Error in Excel				
2	ITEM	QUANTITY			
3	Apple	49			
4	Banana	57		ITEM	QUANTITY
5	Orange	50		Apple	#NAME?
6	Orange	43		Banana	#NAME?
7	Orange	41		Orange	#NAME?
8	Orange	40			
9	Banana	67			
10	Banana	32			

A. The column is not wide enough to accommodate the value
B. There is invalid cell referencing in the formula used
C. The formula uses incorrect argument or operator
D. The reference to a value is not available to a formula
E. The formula references an invalid intersection of cells

(x) In a school registration database, each student must have an ID number that uniquely distinguishes them from everyone else so that duplicate records cannot be created. What is this identifier called?

A. Foreign Key
B. Composite Key
C. Primary Key
D. Candidate Key
E. Alternate Key

2. Study the simplified flowchart below and match each labelled symbol in List A with its appropriate function in List B.



List A	List B
(i) Symbol 1 (ii) Symbol 2 (iii)Symbol 3 (iv)Symbol 4 (v) Symbol 5 (vi)Symbol 6	A. Represents a decision or condition B. Represents the beginning of an algorithm C. Represents input of data D. Represents output of information E. Represents the end of an algorithm F. Represents a processing operation G. Represents data storage H. Represents repetition of an operation

SECTION B (54 MARKS)

Answer **all** questions in this section.

3.

(a) Define the role of an **Internet Service Provider (ISP)** in enabling a school computer lab to connect to the global network.

(b) A local digital media house, *HABARI KWANZA*, is upgrading its computers to run heavy graphic design and spreadsheet applications. Suggest three hardware performance specifications they should check to ensure fast data processing.

(c) You are provided with a Fiber Optic cable, a USB cable, and a Twisted Pair (Ethernet) cable. Identify which cable is best suited to:

(i) Connect a mouse or flash drive over a short distance.

(ii) Transmit high-speed network data across a long distance between towns.

(iii)Set up a local office network connection using standard RJ-45 connectors.

4. (a) Explain the following terms as used in Information and Computer Studies:

(i) Cybercrime
(ii) Data security
(iii) Authentication

(b) Computer malware threatens information systems.
(i) What is malware?
(ii) State any two examples of malware.

5. Most individuals whose regular jobs require conveying messages to different audiences face challenges in communicating their ideas. Sometimes it becomes tough for them to choose whether electronic or non-electronic presentation delivery modes. As IT expert help them to:
(a) Describe four situations in which you would use non-electronic presentations over electronic presentations.
(b) Briefly explain any five qualities of a good presenter.

6. (a) Explain three types of database users found in an organization.

   (b) State the steps followed when creating a new table in a database management system.

7. A Form Four student is using a Desktop Publishing (DTP) application to design a three-column graduation newsletter for Seypalm Secondary School. She notices that text overlapping occurs and some graphics do not sit properly between paragraphs.

   (a) Define the concept of **Text Wrapping** as used in Desktop Publishing and explain how it can solve her design layout problem.

   (b) Differentiate between **Gutter Margin** and **Page Orientation** settings when configuring documents for booklet printing.

   (c) State four quality layout rules the student should maintain to make the newsletter visually appealing and easy to read.

8. Staff at the Dodoma Regional Library complain that it takes a long time to open web pages and share files. An inspection reveals that all computers are connected in a physical Ring configuration, where a single break in the cable drops the entire network, and data packets must pass through every machine.
   (a) Name the problematic network topology currently in use at the library.
   (b) Outline four critical factors a network administrator should consider before upgrading to a different physical network layout.
   (c) Recommend the best alternative network topology to resolve their speed and reliability issues, providing one clear reason for your choice.
   (d) List four essential hardware devices required to install and run the new network layout you recommended.

SECTION C (30 MARKS)

Answer **two questions** in this section

9. There is major concern regarding the production of electronic communication devices and environmental degradation. In four points, explain how the increase in production of electronic communication devices has negative impacts to the environment.

10. Seypalm was telling people that: "Without internet service, life is very difficult in today's world because internet is an integral part of businesses". Discuss the accuracy of Seypalms's statement in six points.

11. A well-designed website is important for organizations and businesses.
    Describe six characteristics of a good website.